

LEASEOS

Live Assist + Unified AI Assistant

Architecture and Build Plan

Version 1.0

19 September 2026

LeaseOS trucking / field operations platform

Purpose

Build one LeaseOS Assistant that can understand voice, LeaseOS context and explicitly shared Android screen content; help with paperwork, trucking, dispatch, mechanics, training and company operations; and carry out only authorized, typed actions through a default-deny authority gateway.

DESIGN STATUS: APPROVED DIRECTION - IMPLEMENTATION PLAN

*This plan extends the existing LeaseOS AI Secretary, provenance, offline-first, role/permission and human-confirmation patterns.
It does not authorize autonomous bypass of compliance, safety, security, device or platform controls.*

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Recommended delivery strategy

Do not start with whole-device control. First build the authority gateway, typed action registry and unified AI Secretary. Then add read-only Live Assist. Only after the evidence, confirmation and policy gates are proven should LeaseOS enable deterministic cross-app actions.

1. Executive build brief

LeaseOS Live Assist is not a separate chatbot and not a generic remote-control bot. It is the visual/voice interaction layer for one unified LeaseOS Assistant. The user talks to one assistant; behind the scenes, an orchestrator routes work to approved capabilities for paperwork, driving, routing, dispatch, compliance, billing, training, mechanics and company operations.

The build must preserve a hard separation between AI reasoning and authoritative system writes. AI can observe, explain, extract, recommend and prepare. Actual business changes pass through a typed Action Gateway that checks tenant, identity, role, device trust, job context, vehicle state, risk, evidence and confirmation requirements before executing.

Primary outcome

A driver, mechanic, dispatcher or office user should be able to say what they need in ordinary language, optionally share the Android screen, and receive help that is grounded in LeaseOS records and approved sources - without giving an LLM direct database write access or unrestricted device control.

Success criteria

- One visible LeaseOS Assistant; specialist routing is internal.
- Works on personal/BYOD Android and company-managed fleet Android devices.
- Hybrid local-first processing: wake phrase, basic OCR/UI understanding and simple commands stay local when practical; cloud models are used only when needed.
- Parked / moving / emergency modes materially change allowed actions, not merely the UI.
- Important answers show source, freshness, confidence and whether content is measured, observed, historical, inferred or confirmed.
- No consequential action is committed without the required authority and confirmation.

- Offline operation degrades safely: cached data remains usable, stale/live dependencies are labeled, writes queue for synchronization.
- Raw screen/audio is transient by default; durable business facts are stored as structured records with provenance.
- Mechanics assistance can diagnose and prepare work, but cannot autonomously declare a vehicle safe or release it to service.

2. Locked design decisions

Decision	Approved direction
Screen scope	Both: LeaseOS-native context plus explicit whole-screen sharing sessions.
Action level	See + understand + operate, but only through typed/approved actions and confirmation gates.
Device deployment	Both BYOD/personal devices and company-managed fleet devices.
AI topology	Hybrid local-first with cloud escalation when required.
Driving safety	Parked / moving / emergency state model.
Assistant UX	One LeaseOS Assistant with automatic specialist routing.
Knowledge	Approved company knowledge + approved external/live sources with provenance.
Permissions	Secure LeaseOS defaults + company-customizable roles; non-bypassable safety floor.
Proactivity	Mixed proactive mode with policy-controlled interruptions and task inbox.
Invocation	Button + voice + optional local wake phrase such as “Hey LeaseOS”.
Memory	Layered memory/retention: session, job, user, equipment, company and audit.
Offline	Graceful degradation with local/cached capabilities and queued synchronization.
Android control	Native APIs/intents first; deterministic accessibility automation only where policy-permitted; no unrestricted autonomous LLM tap loop.

3. Non-negotiable safety and authority rules

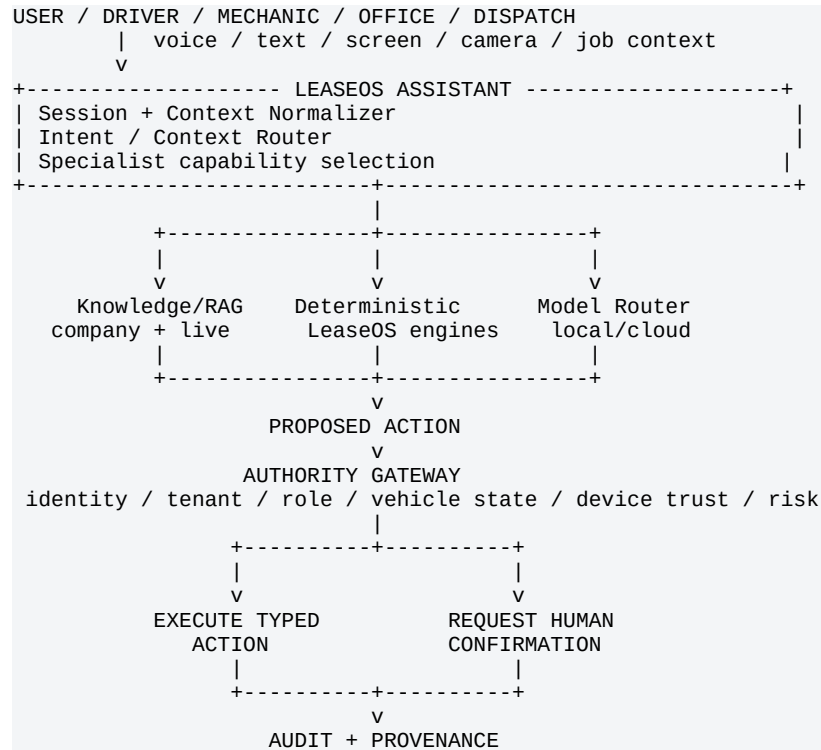
Rule	Implementation meaning
AI does not write directly to authoritative tables	All writes go through typed service/action contracts.
Unknown is not safe	Missing or stale evidence produces UNKNOWN/REVIEW, never a fabricated pass.

Rule	Implementation meaning
Proposal before commitment	OCR, voice extraction, predicted maintenance, billing lines and compliance changes remain proposed until required confirmation occurs.
User authority caps AI authority	The assistant cannot obtain more authority than the authenticated user and current delegated role.
Consequential actions step up	HOS corrections, signatures, billing submission, vehicle release, dispatch acceptance, deletions, external communications and permission changes require explicit policy checks and usually human confirmation.
Driving state is a control input	Moving state disables distracting and nonessential interaction pathways.
Protected content stays protected	LeaseOS does not attempt to bypass Android secure-screen, enterprise, biometric, privacy or app restrictions.
Evidence remains explainable	A user can ask “Why?” and see the records/sources, timestamps, confidence and inference boundary.
Tenant isolation is absolute	No cross-company data leakage through retrieval, chat history, screen observations, embeddings, logs or model context.
Raw capture is temporary by default	Store structured facts and evidence references rather than continuous recordings unless a defined evidence policy requires retention.

Action risk ladder

Level	Examples	Human confirmation
L0 - Observe	Read screen/context; explain; search approved knowledge.	No
L1 - Assist	Highlight, calculate, guide, draft.	No
L2 - Prepare	Populate forms, work-order drafts, dispatch/invoice drafts.	Usually no to save draft
L3 - Routine execute	Reversible, approved actions such as navigation handoff, filters, draft save, attach evidence.	Policy dependent
L4 - Consequential	HOS correction, billing submit, dispatch acceptance, safety declaration, vehicle release, signature, delete, external send.	Yes
L5 - Restricted	Bypass compliance/safety/security, fabricate evidence, impersonate signature, weaken controls, mark unknown as verified.	Never

4. Target architecture



Core logical modules

Module	Responsibility
Assistant Session Manager	Starts/ends assistant sessions; tracks user, device, job/unit, vehicle state, connectivity and allowed scopes.
Intent/Context Router	Routes a request to paperwork, routing, HOS, mechanics, training, dispatch, billing, company operations or a combination.
Capability Registry	Declares what each capability can read, prepare or execute; inputs, outputs, risk level, confirmation, offline availability.
Authority Gateway	Default-deny enforcement point for every assistant-initiated action.
Action Registry	Typed action vocabulary such as OPEN_JOB, NAVIGATE_TO, CREATE_DEFECT, PREPARE_FIELD_TICKET, CHECK_HOS and CREATE_WORK_ORDER.
Knowledge Gateway	Retrieves approved company knowledge, LeaseOS records, authoritative regulatory/OEM material and live external data with source metadata.
Model Router	Chooses local rules/model vs cloud model based on task complexity, privacy, connectivity and policy.
Live Assist Capture Engine	Explicit Android MediaProjection session, frame-change detection, local OCR/UI parsing and redaction.
Platform Adapters	Android intents/deep links, 511/map integrations, email/messaging, OEM/telematics and other official APIs.

Module	Responsibility
Task Inbox	Turns unresolved assistant work into durable operational tasks.
Audit/Provenance Ledger	Records source, confidence, actor, approver, action, before/after, device/session and result.

5. Capability model and user experience

5.1 One assistant, specialist capabilities

Capability	Scope
Paperwork	Voice/screen/camera extraction, gap detection, read-back, proposed forms, tickets, manifests, inspections and billing evidence.
Driver/Trucking	Job instructions, routes, offline maps, HOS assistance, inspections, load/stop context, road restrictions and contacts.
Dispatch	Assignment context, readiness, route/status explanation, job contacts, exceptions and queued decisions.
Mechanics	Defects, diagnostic context, work orders, unit history, sensor readings, parts/labor/evidence and guided troubleshooting.
Safety/Compliance	Explains blocks, missing evidence and rule state; never invents compliance or bypasses hard gates.
Billing/Office	Prepared invoices, evidence completeness, rejected/missing ticket detection, customer requirements and review queues.
Training/How-to	LeaseOS usage, company SOPs, approved training, equipment procedures and escalation to authoritative manuals.
Company Operations	Employee/unit/document status, task queues, expiries, client requirements, summaries and authorized administration.

5.2 Context-sensitive behavior

State	Allowed behavior
Parked	Full voice and visual assistance; prepare/review forms; routine approved actions; L4 confirmation as required.
Moving	Voice-first; navigation, route/HOS/safety information, read-aloud, alerts and task queuing. Complex typing/review and nonessential cross-app interaction disabled.
Emergency/Critical	Surface coordinates, emergency contacts/ERP/ERAP, critical instructions and approved communication shortcuts. Emergency mode does not bypass compliance/security controls.
Offline	Use cached map/company/manual/job data; local extraction/rules where available; label stale dependencies; queue sync/actions requiring connectivity.

5.3 Explainability

Operationally meaningful answers should support a “Why?” view. The answer payload should carry source class, source identifier, observed/verified time, freshness, confidence and the boundary between evidence and model inference.

```
Example diagnostic statement
MEASURED    coolant_temp = 112 C      source: ECU sensor
OBSERVED    residue near upper hose   source: mechanic photo
HISTORICAL  similar loss 18 days ago  source: W0-1842
INFERRED    probable cooling leak      source: assistant reasoning
CONFIRMED   upper radiator hose leak  verified_by: mechanic
```

6. Android Live Assist architecture

6.1 BYOD/personal devices

- Use Android MediaProjection for explicit whole-screen capture sessions. User consent is requested for each capture session where required by Android behavior.
- Show a persistent LeaseOS “LIVE ASSIST ACTIVE” control plus Android/system capture indication. Either stop path terminates capture and releases resources.
- Process frame fingerprints, UI semantics and OCR locally first; do not continuously stream video to the cloud.
- Redact sensitive regions when possible before cloud escalation. Treat password/PIN/biometric/secure screens as user-only steps.
- Use official APIs, LeaseOS APIs, Android intents and deep links before any accessibility-based automation.
- If AccessibilityService is used for automation, constrain it to declared, narrow, deterministic workflows. Do not give an LLM an unrestricted plan-and-tap loop.

6.2 Company-managed fleet devices

- Use Android Enterprise/MDM policy to standardize allowed apps, security posture, data-sharing policy and Live Assist availability.
- Respect device-owner/profile-owner restrictions that disable content capture or AI/agent automation.
- Keep company policy separate from the non-bypassable LeaseOS safety floor: MDM may reduce capability, never silently expand assistant authority.
- Support separate profiles/policies for driver tablets, shop/mechanic devices and office/dispatch devices.

6.3 Capture pipeline

```
MediaProjection frame
|
v
local fingerprint / change detector
|---- no meaningful change ----> discard
v
semantic UI extraction + OCR + local classifier
|
+---- enough context -----> structured observation
|
v
redaction + minimum-context package
|
v
cloud vision/reasoning only when needed
|
v
proposed fact/action -> Authority Gateway
```

6.4 Android/Google Play implementation constraints

Current platform constraint - re-verify before store submission

Google Play policy permits narrow, clearly understood deterministic accessibility automation, but prohibits using AccessibilityService so an app autonomously initiates, plans and executes actions or decisions. Build LeaseOS around typed actions, intents/APIs and deterministic adapters - not unrestricted AI UI control.

Area	Build requirement
MediaProjection	Request user consent for capture; use required foreground service type/permission; handle projection stop callback and release surfaces/resources.
AccessibilityService	Prominent disclosure/consent and Play declaration if used; narrowly scoped deterministic workflows; prefer narrower APIs whenever possible.
Secure/protected screens	Honor platform/app capture restrictions. Do not attempt bypasses.
Enterprise restrictions	Detect and honor device policy restrictions such as content-capture or AI automation controls.
Third-party AI	Data-safety/user-data responsibilities remain with LeaseOS; minimize data and disclose processing appropriately.

7. Knowledge, memory, provenance and offline behavior

7.1 Source classes

Source class	Use
AUTHORITATIVE_REGULATORY	Government/regulator rule or dataset.
OEM_TECHNICAL	Manufacturer/service documentation.
COMPANY_POLICY	Approved employer SOP, customer requirement or internal policy.
LEASEOS_RECORD	Operational record already in LeaseOS.
LIVE_EXTERNAL	Current road/511/weather/partner API response.
AI_GENERAL_KNOWLEDGE	Explanation only; never sufficient evidence for a consequential operational fact.

7.2 Layered memory

Layer	Retention/use
Session	Temporary conversation/Live Assist context; short TTL by default.
Job	Current assignment, customer, route, unit/trailer, forms, contacts, evidence and unresolved tasks.

Layer	Retention/use
User	Approved role, accessibility, units/preferences and non-sensitive workflow preferences.
Equipment	Defects, maintenance, sensor events, inspections, parts and recurring issues.
Company	Approved SOPs, forms, training, customer rules, rates and knowledge.
Audit	Immutable evidence of consequential actions, approvals and outcomes; not ordinary chatbot memory.

7.3 Offline-first behavior

Connectivity state	Behavior
FULL	Local + cloud AI, live sources, online LeaseOS services.
LIMITED	Local AI/rules + cached LeaseOS data; cloud/live work queued or selectively retried.
OFFLINE	Local voice commands where supported, cached knowledge/maps/manuals, paperwork drafting, photo/note evidence, queued sync.
STALE/UNKNOWN	Anything requiring fresh external verification is clearly marked and cannot silently satisfy a compliance/safety gate.

8. Data model and service contracts

Names below are recommended logical contracts. Adapt them to the existing LeaseOS schema and naming conventions instead of creating parallel concepts.

8.1 Recommended tables/entities

Entity	Key content
assistantSessions	tenantId, userId, deviceId, jobId/unitId, startedAt, endedAt, mode, drivingState, scopes, connectivityState
assistantTasks	type, context refs, owner/role, priority, status, dueAt, sourceSessionId, blockingReason
assistantActions	actionType, argsHash, actor, actingRole, riskLevel, state, result, reversible, timestamps
assistantApprovals	actionId, requiredRole, approver, method, decision, reason, timestamp
assistantEvidenceRefs	action/task/message ref -> source class/id/version, observedAt, confidence, freshness, excerpt/hash
liveAssistSessions	projection/session state, allowed capture scopes, retention policy, start/stop reason, policy snapshot

Entity	Key content
screenObservations	ephemeral/short-lived structured observation; sourceApp, UI element/text, confidence, frame hash, redaction state
capabilityDefinitions	capability/action metadata: input schema, output schema, risk, offline support, required scopes, confirmation
assistantPolicies	tenant overrides within LeaseOS hard limits: proactivity, retention, allowed adapters, cloud escalation, role mappings
knowledgeSources	source identity, authority class, version, jurisdiction/customer/OEM scope, verifiedAt, license/usage constraints
assistantAuditEvents	append-only assistant lifecycle/action/approval/security events linked to existing audit ledger

8.2 Provenance type

```
Provenance {
  sourceClass, sourceId, sourceVersion,
  observedAt, verifiedAt, verifiedBy,
  confidence, freshness, status,
  precision?, evidenceHash?
}
```

status: pending | confirmed | rejected | unknown | stale

8.3 Typed action contract

```
ActionDefinition {
  id: "CREATE_WORK_ORDER",
  inputSchema,
  allowedRoles,
  allowedDrivingStates,
  riskLevel: "L2",
  confirmationPolicy,
  requiredEvidence,
  reversible,
  offlinePolicy,
  executorAdapter
}
```

8.4 Initial action vocabulary

- OPEN_JOB, OPEN_ROUTE, OPEN_UNIT, SHOW_DOCUMENT, CALL_JOB_CONTACT
- NAVIGATE_TO, CHECK_ROUTE, CHECK_HOS, CHECK_DISPATCH_READINESS
- CREATE_DEFECT, CREATE_WORK_ORDER, ADD_WORK_ORDER_EVIDENCE, PREPARE_UNIT_RELEASE
- ADD_LOAD, ADD_UNLOAD, ADD_DISPOSAL_TICKET, ATTACH_PHOTO, ATTACH_DOCUMENT
- PREPARE_FIELD_TICKET, PREPARE_MANIFEST, PREPARE_BILLING, REQUEST_SIGNATURE
- OPEN_ERP, SHOW_EMERGENCY_CONTACTS, QUEUE_TASK, EXPLAIN_BLOCK, SHOW_WHY

9. Implementation phases and checkpoints

Sequencing rule

Build authority, provenance and typed actions before adding screen control. Build read-only screen understanding before cross-app automation. Treat every checkpoint as a shippable safety boundary.

P0 - Baseline and contracts

Scope	Deliverables	Exit gate
Inventory existing AI Secretary, audit, role, offline, mobile and routing contracts; freeze duplicate concepts; define module ownership and feature flags.	Architecture map; ADRs; dependency map; feature flags; policy snapshot tests.	No new AI write path is allowed until P1.

P1 - Authority Gateway + Action Registry

Scope	Deliverables	Exit gate
Implement default-deny action evaluation, L0-L5 risk, role/tenant/device/job/driving-state checks, approvals and audit receipt.	Typed action registry; authorization evaluator; approval workflow; audit schema.	Every assistant write must fail closed when policy context is missing.

P2 - Unified AI Secretary / Orchestrator

Scope	Deliverables	Exit gate
Unify paperwork and specialist routing behind one session/orchestrator; schema-constrained extraction; minimum-question gap detection; read-back.	Session API; intent router; capability registry; structured proposals.	No direct model-to-DB mutation; all proposed fields carry provenance.

P3 - Knowledge + Provenance Gateway

Scope	Deliverables	Exit gate
Build source classes, retrieval filters, freshness/authority metadata, "Why?" payloads and tenant isolation.	KnowledgeSource registry; provenance adapter; source/freshness UI contract.	Important answer can enumerate evidence and inference boundary.

P4 - Voice + driving-state shell

Scope	Deliverables	Exit gate
Button/mic flows, optional local wake phrase, parked/moving/emergency behavior, visible mic/session controls.	Voice session manager; driving-state policy hook; command queue.	Moving-mode test suite proves distracting actions are unavailable.

P5 - In-app Live Assist

Scope	Deliverables	Exit gate
Assistant understands LeaseOS UI state directly without whole-screen capture; highlights and guides users; opens typed LeaseOS actions.	Screen/context contract inside LeaseOS; help overlays; how-to routing.	Can complete top LeaseOS help workflows without MediaProjection.

P6 - Whole-screen read-only Live Assist

Scope	Deliverables	Exit gate
Add MediaProjection session, local frame-change detection, OCR/UI parsing, redaction and cloud escalation. No cross-app execution yet.	Android capture service; observation pipeline; session permission UI; stop lifecycle.	Capture always explicit/visible; revocation instantly stops processing; raw frames obey retention.

P7 - Deterministic external action adapters

Scope	Deliverables	Exit gate
Add native intents/deep links/APIs and tightly scoped deterministic accessibility actions only where needed.	Adapter SDK; Maps/navigation adapter; approved messaging/email handoff; selected app workflows.	No arbitrary tap loop; each action maps to a reviewed ActionDefinition.

P8 - Hybrid model router + offline assistant

Scope	Deliverables	Exit gate
Local/simple task routing, cloud escalation policy, cached knowledge/manuals, queued assistant work and sync conflict handling.	Model router; offline capability manifest; encrypted queue; stale-state UX.	Loss of network does not convert unknown data into verified results.

P9 - Managed fleet mode

Scope	Deliverables	Exit gate
Android Enterprise/MDM policy integration, device posture and managed restrictions.	Managed policy reader; fleet admin settings; device compliance indicators.	Enterprise restrictions always reduce/disable capability safely.

P10 - Mechanic + company operations expansion

Scope	Deliverables	Exit gate
Mechanic diagnostics/work orders plus office/dispatch/billing/training capabilities using the same contracts.	Mechanic context resolver; equipment evidence; operations capability modules.	Mechanic assistant never independently closes safety-critical release decision.

P11 - Proactive assistant + task inbox

Scope	Deliverables	Exit gate
Policy-controlled alerts, missing-evidence detection, billing-ready queue, expiring docs, defects and unresolved assistant tasks.	Task rules; notification policy; task inbox UI; escalation.	No alert can auto-escalate authority or auto-commit L4 actions.

P12 - Security/privacy/policy hardening

Scope	Deliverables	Exit gate
Threat model, Play declarations, disclosure/consent, data minimization, log redaction, abuse tests, retention controls.	Policy documentation; privacy UX; pen-test findings; store submission checklist.	Zero known critical security/privacy issues; declarations match actual behavior.

P13 - Field pilot and production gate

Scope	Deliverables	Exit gate
Real-device testing across supported Android versions, weak/offline networks, personal and managed devices, driver/shop/office pilots.	Pilot report; telemetry; defect burn-down; rollback plan; release runbook.	Production gates in Section 10 all green.

Recommended release checkpoints

Checkpoint	Phases	Capability
Checkpoint A - Assistant Core	P0-P3	One assistant, provenance, typed actions, no screen sharing.
Checkpoint B - Voice/In-app	P4-P5	Voice + in-app guided assistance, driving-state controls.
Checkpoint C - Live Assist Observe	P6	Whole-screen understanding only; no autonomous cross-app execution.
Checkpoint D - Controlled Actions	P7-P9	Approved adapters, offline/hybrid behavior and managed fleet mode.
Checkpoint E - Operational Suite	P10-P11	Mechanics/company operations/proactive tasks.
Production Candidate	P12-P13	Security/policy/device/pilot gates complete.

10. Test strategy and production gates

10.1 Required automated test families

- Authorization matrix: every action x role x tenant x driving state x risk level.
- Tenant isolation: retrieval, embeddings, screen observations, task queues and audit refs.
- Proposal/confirmation: uncertain OCR/voice values never silently become confirmed records.
- Capture lifecycle: deny, start, rotate, app switch, protected content, revoke, process death, reboot.
- Moving-mode enforcement: prohibited UI/action pathways remain unavailable even if requested by model output.
- Offline/stale handling: cached data labels, queued writes, retries, conflict resolution and no false verification.
- Action idempotency/replay: duplicate model/tool calls cannot duplicate invoices, tickets, messages or work orders.
- Model failure/hallucination: invalid/unknown action or argument is rejected before executor.

- Approval step-up: L4 action cannot execute with missing/expired/wrong-role approval.
- Managed policy: content-capture/AI restrictions are detected and honored.
- Retention/redaction: transient screen frames/audio are removed according to policy and sensitive values are not leaked to logs.
- Accessibility and one-handed/mobile: large text, screen reader, gloves/large touch targets where supported, voice fallback.

10.2 Production gates

Gate	Production requirement
G1	0 direct LLM writes to authoritative LeaseOS business tables.
G2	100% assistant actions map to registered typed ActionDefinitions.
G3	100% L4 actions show required confirmation and audit receipt.
G4	Unknown/stale evidence cannot produce PASS/SAFE/VERIFIED.
G5	All important operational answers expose source + time/freshness + confidence/inference status.
G6	Capture sessions require explicit current consent/visibility and stop cleanly on revocation.
G7	No unrestricted autonomous AccessibilityService planning/execution path exists in Play-distributed build.
G8	Tenant isolation and authorization tests pass for every assistant data path.
G9	Offline behavior is safe and understandable; queued records are durable and encrypted.
G10	Security/privacy review has no unresolved critical/high issues affecting production use.
G11	Field pilot covers personal + managed Android, moving/parked states, weak network and real paperwork/mechanic workflows.
G12	Rollback/kill switch can disable Live Assist or specific adapters without disabling core LeaseOS.

11. Security, privacy and platform compliance

11.1 Threat model priorities

Threat	Mitigation
Prompt injection from screen/doc	Treat external content as untrusted data. It cannot redefine assistant policy or authorize actions.
Cross-tenant retrieval	Tenant-scoped indexes/queries and policy-enforced joins; tests for every retrieval path.
Credential capture	Detect/redact sensitive UI where possible; never persist passwords/PINs/tokens from capture; secure screens remain user-only.

Threat	Mitigation
Over-broad screen retention	Frame TTL and structured-observation model; evidence retention must be explicit and job/policy linked.
Unauthorized external action	Typed allowlist + risk/role/context checks + idempotency + confirmation.
Model tool abuse	Model can request only known capabilities; gateway validates all arguments server-side.
Lost/stolen device	Device binding, secure local storage, session timeout, revocation, managed wipe where applicable.
Audit tampering	Append-only audit trail with integrity controls and durable references to evidence/action receipts.
Proactive distraction	Moving-mode suppression and policy-controlled interrupt classes.
Stale regulatory/live data	Freshness thresholds and explicit UNKNOWN/STALE state; no silent fallback to general model knowledge.

11.2 Privacy defaults

- Whole-screen sharing is opt-in, session-scoped and visibly active.
- Raw screen/video/audio is not a general historical archive.
- Cloud escalation transmits the minimum context needed for the task and records source/processing policy.
- Company administrators can shorten optional AI-memory retention but cannot erase legally/operationally required audit records through the conversational assistant.
- User-facing permission dashboard shows current screen, mic, location, cloud and external-action access.
- Assistant logs must avoid secrets and unnecessary personal content.

12. Pilot and rollout

12.1 Recommended pilot cohorts

Cohort	Size	Focus
Driver pilot	5-10 drivers	Voice, parked/moving controls, paperwork preparation, in-app help, read-only Live Assist, route handoff.
Shop pilot	2-5 mechanics	Defect/work-order flows, photos, unit history, OEM/company knowledge, diagnostic explanations.
Office/dispatch pilot	3-5 users	Billing readiness, missing evidence, dispatch exceptions, task inbox, role/approval behavior.
Managed-device pilot	Small fleet subset	MDM policy, capture restrictions, device trust, remote policy change and rollback.

12.2 Pilot metrics

- Paperwork completion time and number of fields/questions eliminated.
- Assistant proposal acceptance/rejection rate by field and workflow.

- False-positive/false-confidence incidents, especially safety/compliance/mechanics.
- Number of times moving-mode correctly deferred a distracting/consequential task.
- Cloud calls, bandwidth, battery impact and local-vs-cloud routing ratio.
- Capture-session abandon/revoke rate and privacy complaints.
- Missing-evidence reduction before billing/dispatch.
- Mechanic diagnostic usefulness without increasing unsafe release behavior.
- Offline queue success, sync conflict rate and time-to-recovery.
- Support requests resolved by LeaseOS How-To without human escalation.

13. Recommended backlog

ID	Area	Task	Phase
LA-001	Architecture	Map existing AI/role/audit/offline modules; identify reuse and duplicates.	P0
LA-002	Contracts	Create shared Provenance contract and status vocabulary.	P0
LA-003	Security	Create AssistantAction + Approval receipt contract.	P1
LA-004	Security	Implement default-deny Authority Gateway.	P1
LA-005	Actions	Build Action Registry with first 15 LeaseOS actions.	P1
LA-006	AI	Create unified assistant session/orchestrator.	P2
LA-007	AI	Connect schema-driven paperwork engine and read-back.	P2
LA-008	Knowledge	Implement source-class/freshness/provenance retrieval.	P3
LA-009	UX	Build “Why?” evidence panel contract.	P3
LA-010	Voice	Add push-to-talk + optional local wake phrase.	P4
LA-011	Safety	Implement driving-state action matrix.	P4
LA-012	UX	Build in-app Assist overlay/help mode.	P5
LA-013	Android	Implement MediaProjection service + explicit consent lifecycle.	P6
LA-014	Android	Implement frame fingerprint/OCR/redaction pipeline.	P6
LA-015	Android	Build Live Assist permission/session dashboard.	P6

ID	Area	Task	Phase
LA-016	Adapters	Create adapter SDK and navigation intent adapter.	P7
LA-017	Adapters	Add deterministic accessibility runner only for reviewed narrow workflows.	P7
LA-018	Offline	Build model router, local capability manifest and queued assistant actions.	P8
LA-019	Enterprise	Read/enforce Android Enterprise restrictions and device policy.	P9
LA-020	Mechanics	Build unit/defect/work-order mechanic context resolver.	P10
LA-021	Ops	Add billing/dispatch/training/company-operation capability definitions.	P10
LA-022	Tasks	Build assistant task inbox + proactive rule engine.	P11
LA-023	Privacy	Implement retention/redaction policy tests and disclosure UX.	P12
LA-024	Compliance	Prepare Play Accessibility/MediaProjection/Data Safety submission evidence as applicable.	P12
LA-025	QA	Real-device matrix + pilot runbook + rollback/kill switch.	P13

14. Source alignment and reference notes

14.1 Existing LeaseOS alignment

This plan intentionally extends, rather than replaces, existing LeaseOS patterns found in the project material:

- LEASEOS_BUILD_PLAN-3: a single schema-driven conversational form engine; per-field value/precision/source/confidence; minimum missing questions; mandatory read-back; commit with provenance; automation proposes and humans confirm before authoritative commitment.
- LeaseOS prototype role model: Driver, Dispatcher, Mechanic, Office, Manager and Administrator with distinct visibility/change rights.
- LeaseOS offline prototype: encrypted local storage, queued events/photos and explicit last-sync/offline status.
- LeaseOS routing evidence design: unknown is not treated as clear; decisions retain evidence sufficient to reproduce/explain the verdict.

14.2 Android / Google Play references checked for this plan

Reference	URL
Google Play - Use of the AccessibilityService API	https://support.google.com/googleplay/android-developer/answer/10964491?hl=en
Android Developers - Foreground service types / mediaProjection	https://developer.android.com/develop/background-work/services/fgs/service-types
Android 14 behavior - user consent for each MediaProjection capture session	https://developer.android.com/about/versions/14/behavior-changes-14
Android media capture / MediaProjection	https://developer.android.com/media/platform/av-capture
Android UserManager - DISALLOW_CONTENT_CAPTURE	https://developer.android.com/reference/android/os/UserManager#DISALLOW_CONTENT_CAPTURE
Android Enterprise - What is new in Android 17 / agentic automation controls	https://developer.android.com/work/versions/android-17
Google Play - July 15, 2026 policy announcement / third-party AI data responsibility	https://support.google.com/googleplay/android-developer/answer/17134731?hl=en

Revalidation note

Android APIs, Google Play policy and enterprise controls change. Re-check current official documentation at the start of P6, again at P12, and immediately before Play Store/enterprise distribution submission.

Implementation handoff

The next design stage should define the Assistant Brain / Agent Orchestrator in detail: intent classification, specialist capability composition, tool/action selection, context budgeting, conflict resolution, evidence ranking, retries/fallbacks and deterministic handoff into the Authority Gateway. That orchestrator should be built on top of P0-P3 contracts, not before them.